COMPASS_SynopticCB_PW_DOC_202511_Data

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

```
## Setup Information
```

```
###### Run information - PLEASE CHANGE
  Date_Run = "11/10/25"  #Date that instrument was run
  Run_by = "Stephanie J. Wilson"  #Instrument user
  Script_run_by = "Stephanie J. Wilson" #Code user
  run_notes = "One sample from SWH was not refridgerated and has been removed from the analysis in this
  samples <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names
  samples_pattern <- paste(samples, collapse = "|")
    #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
  chks_name = "Chk_Std_50C_2N"  #what did you name your check standards?

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
    raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202511.txt"

#file path and name for raw all peaks file
    raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202511_allpeaks.txt"

#file path and name of processed data file
    processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202511.csv"


###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"

#qaqc log file path for this year
  Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAQClog_2025.csv"
```

##Set Up Code

##Read in metadata and create similar sample IDs for matching to samples

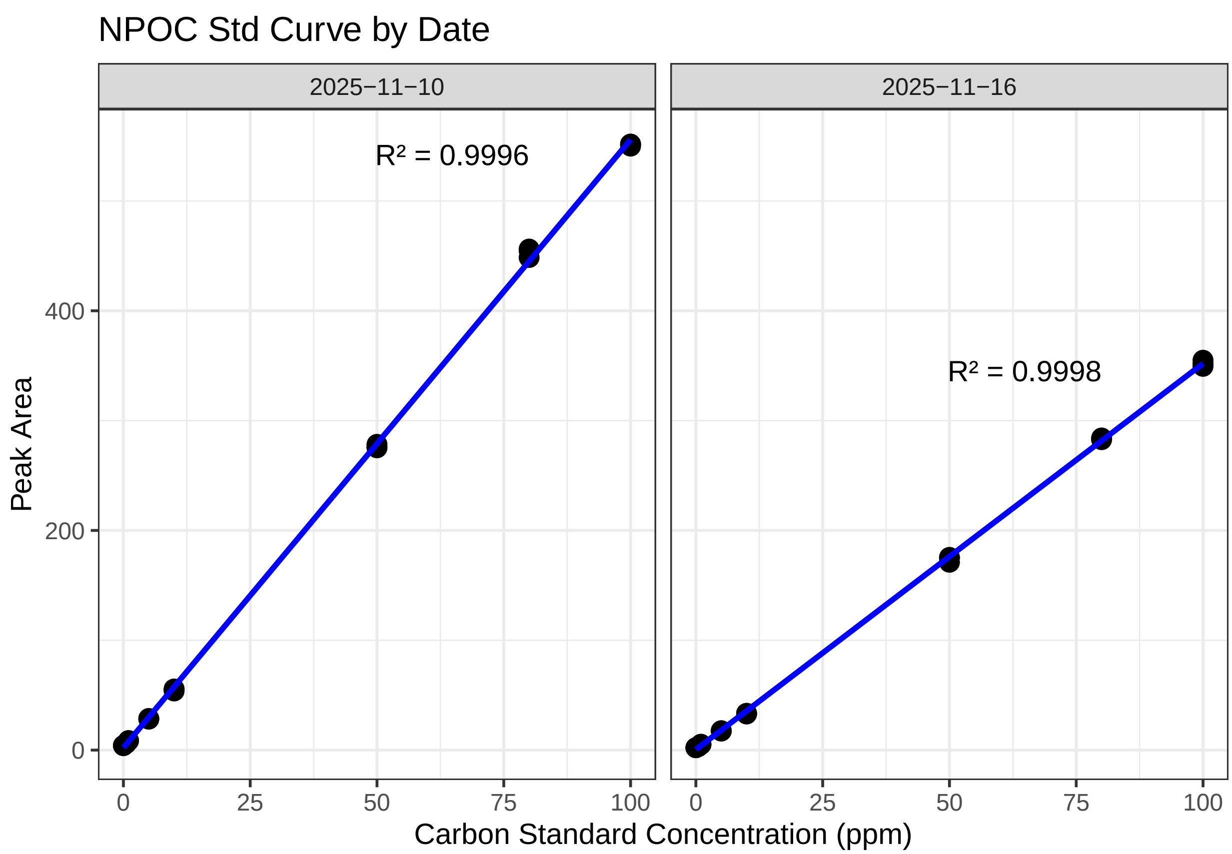
Import Data Functions

Import Sample Data

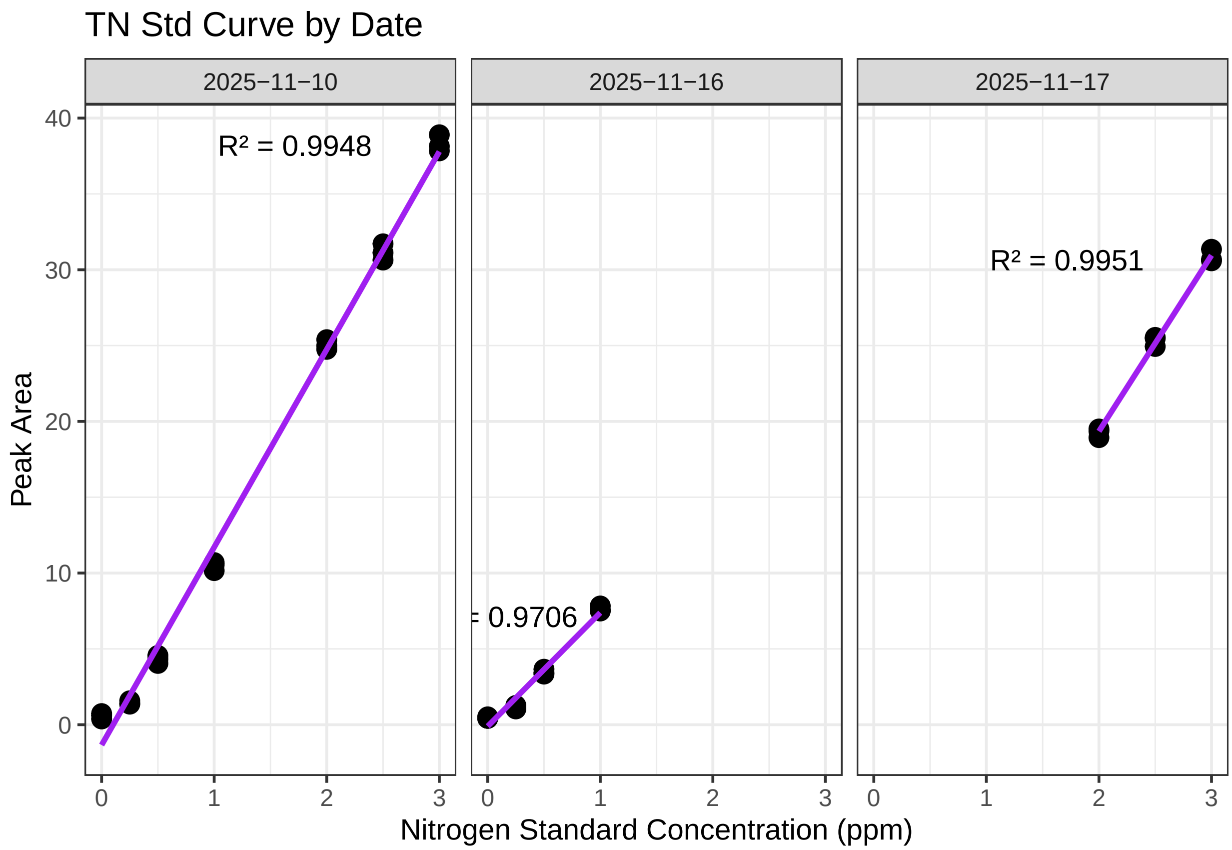
```
## Import Sample Data

## New names:
## * ‘‘ -> ‘...14‘

## # A tibble: 6 x 4
##   sample_name              npoc_raw tdn_raw run_datetime
##   <chr>                       <dbl>   <dbl> <chr>
## 1 202511_SWH_UP_LysA_10cm      7.69   0.272 11/10/2025 8:55:45 PM
## 2 202511_SWH_UP_LysA_20cm      7.94   0.259 11/10/2025 9:27:08 PM
## 3 202511_SWH_UP_LysA_45cm     15.0    0.329 11/10/2025 9:58:24 PM
## 4 202511_SWH_UP_LysB_10cm     13.9    0.458 11/10/2025 10:30:24 PM
## 5 202511_SWH_UP_LysB_20cm      8.93   0.333 11/10/2025 11:01:29 PM
## 6 202511_SWH_UP_LysB_45cm     30.3    0.696 11/10/2025 11:33:28 PM
```

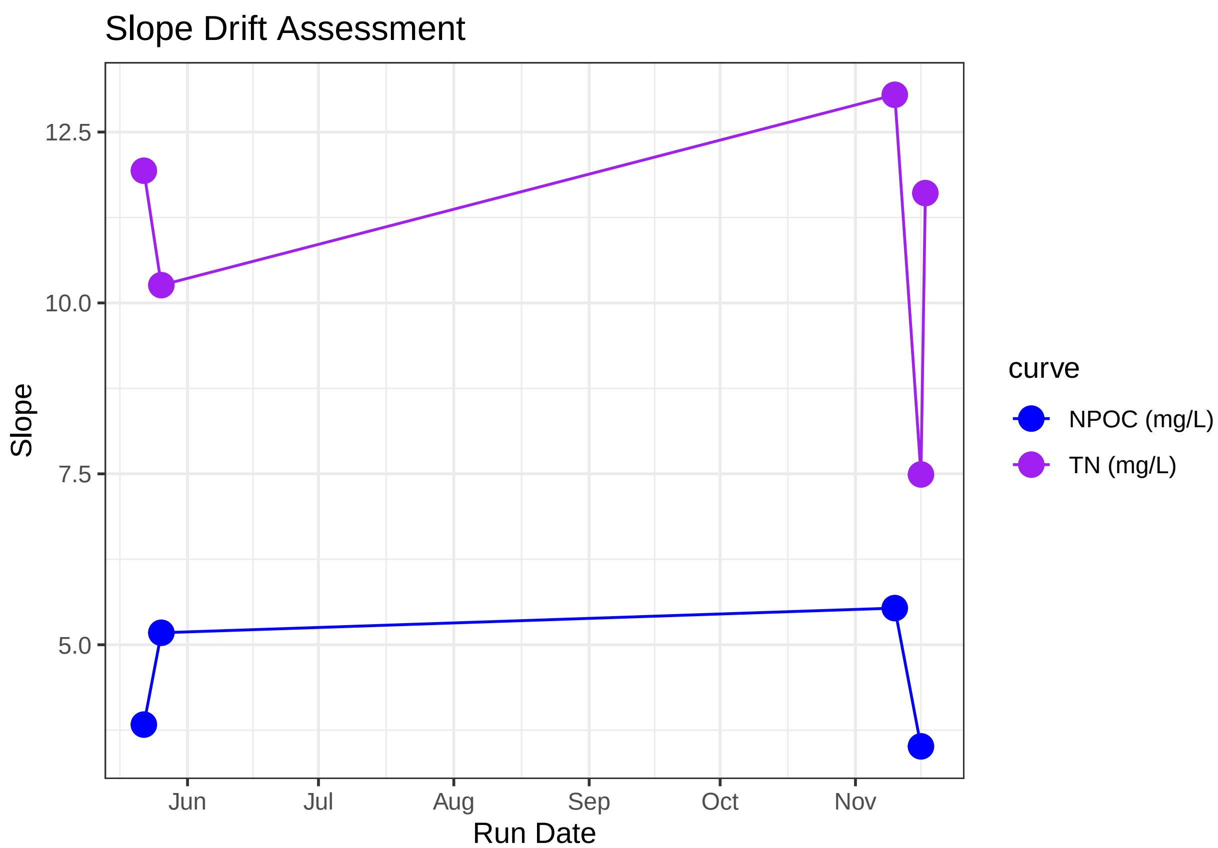
Assessing Standard Curves

```
## Assess the Standard Curves

## New names:
## ‘geom_smooth()‘ using formula = ’y ~ x’
## * ‘‘ -> ‘...18‘
```



```
## `geom_smooth()` using formula = 'y ~ x'
```






```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

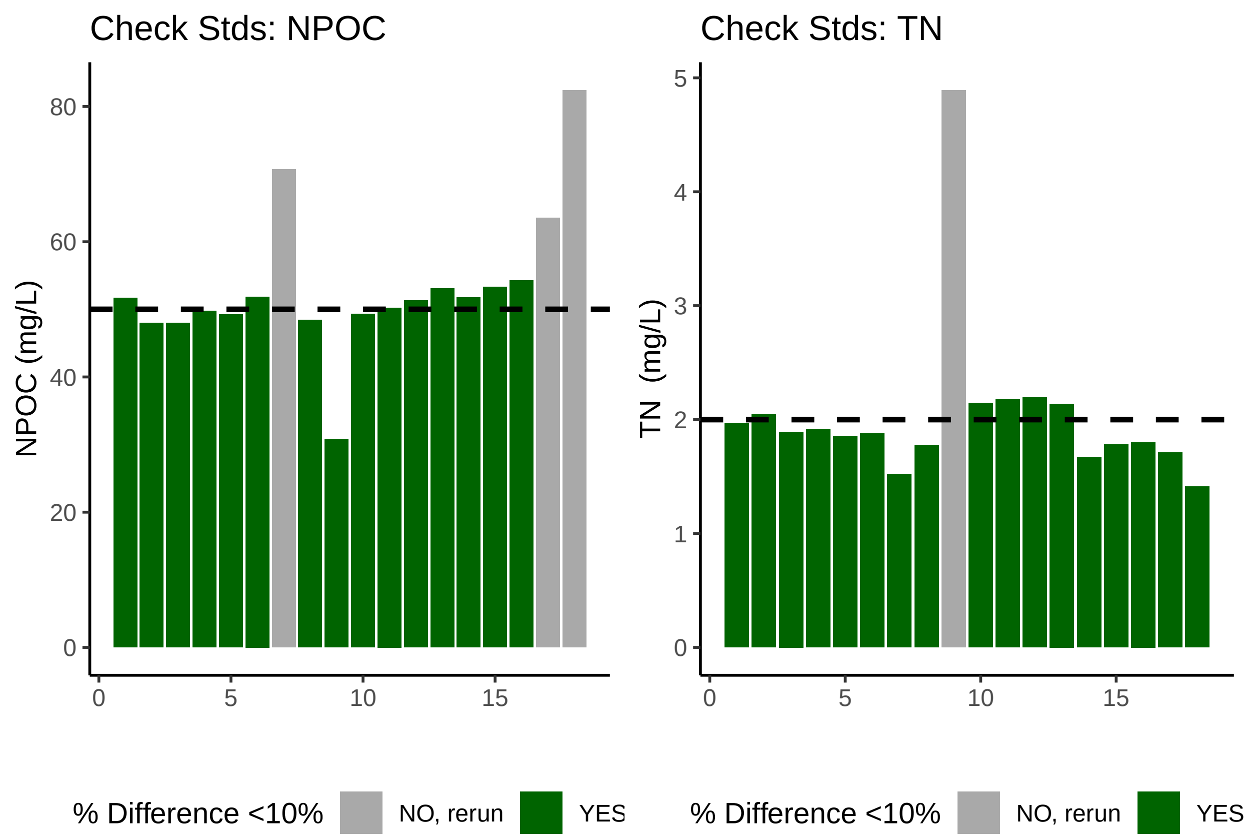
Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * `` -> `...14`
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```




```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

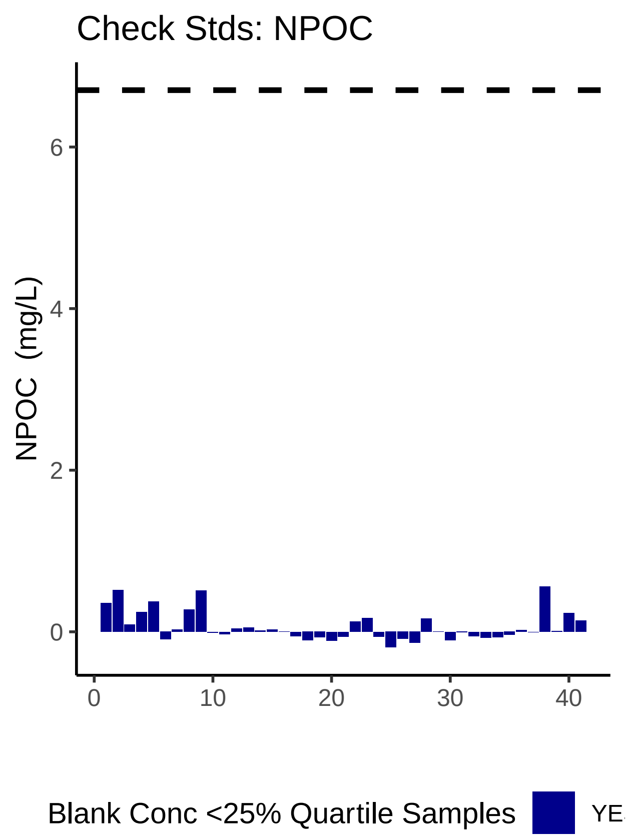
Assess Blanks

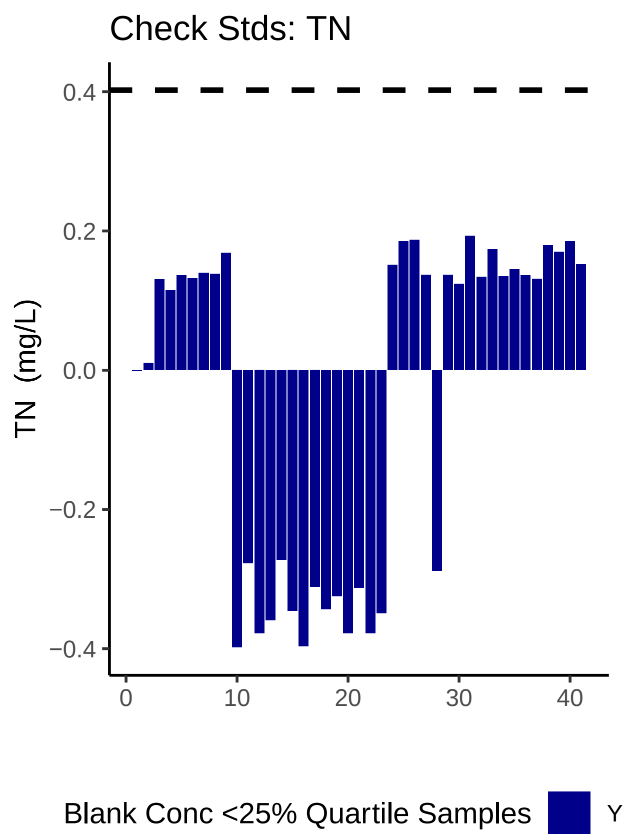
```
## Assess Blanks
```

```
## New names:
## * ‘‘ -> ‘...14‘
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```

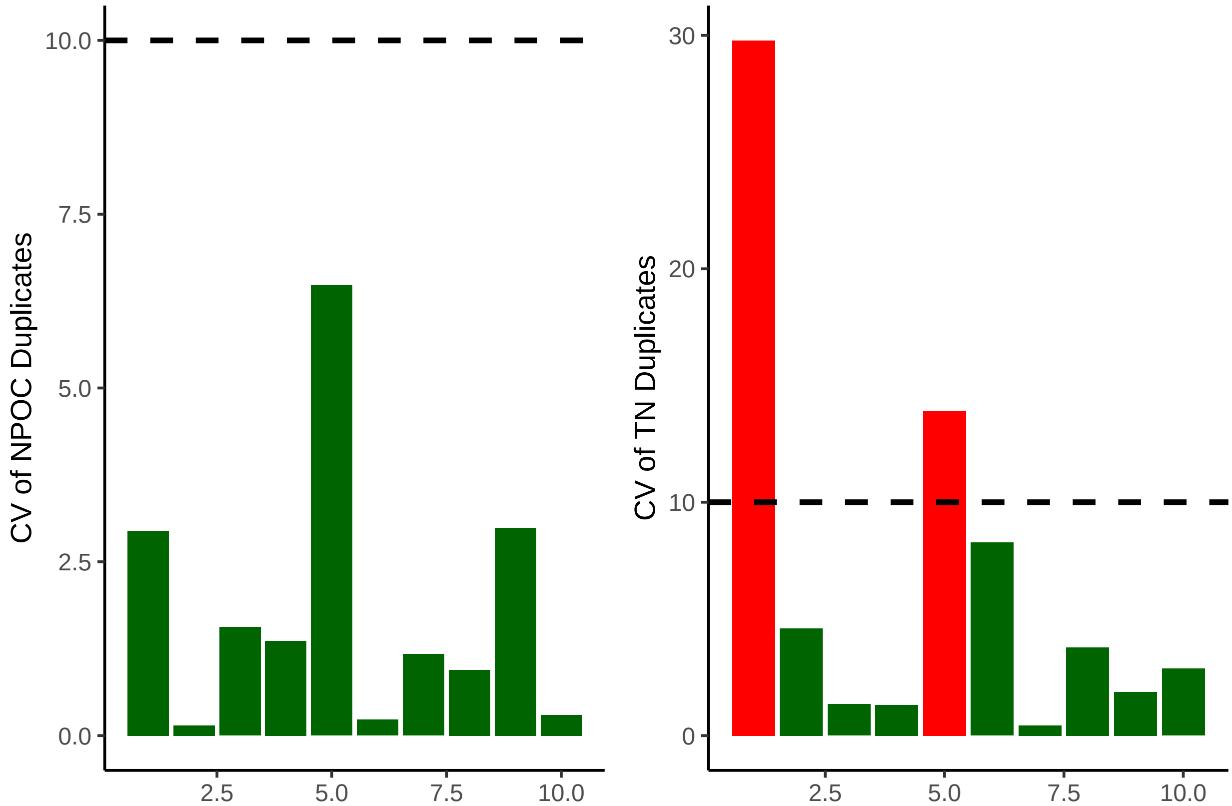





```
## carbon blanks:

## [1] 0.06378463

## nitrogen blanks:

## [1] -0.03622024
```

Assess Duplicates

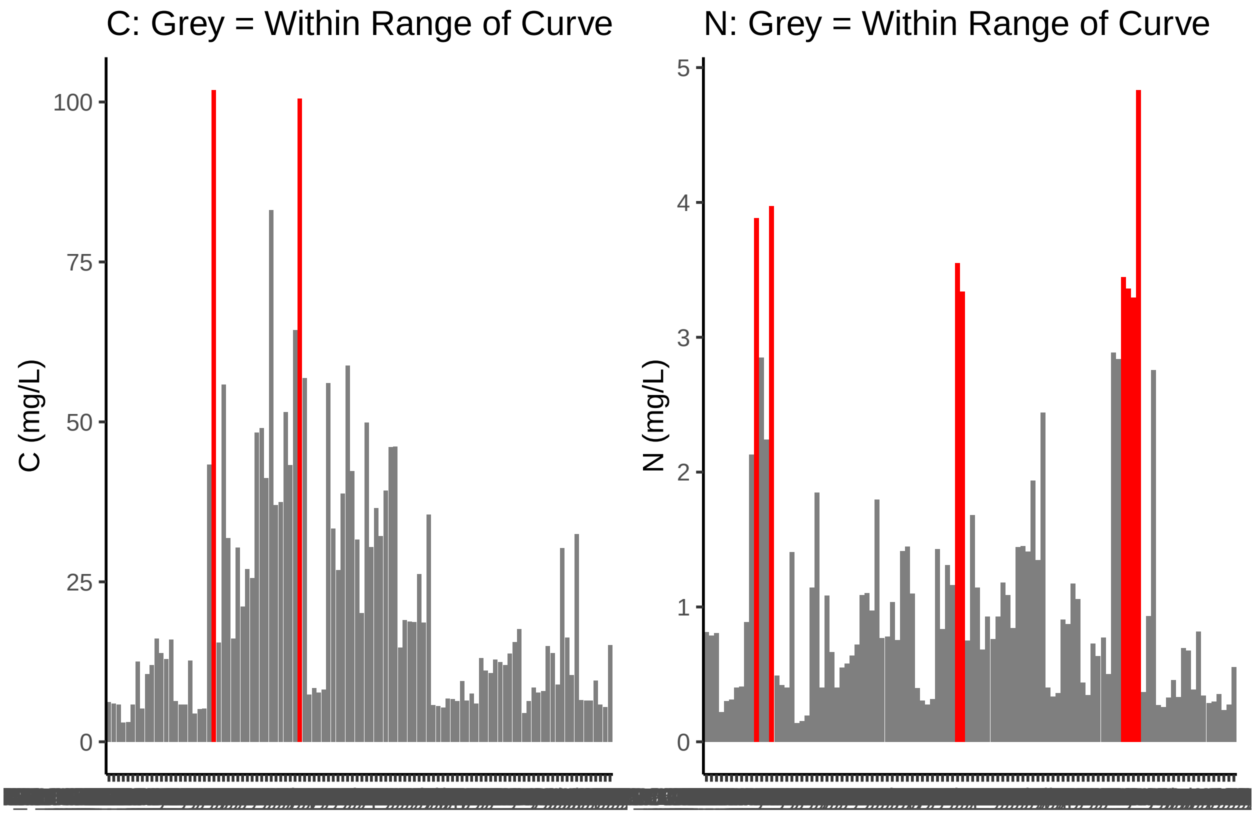
```
## Assess Duplicates

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```




```
## [1] ">60% of Carbon Duplicates have a CV <10%"

## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

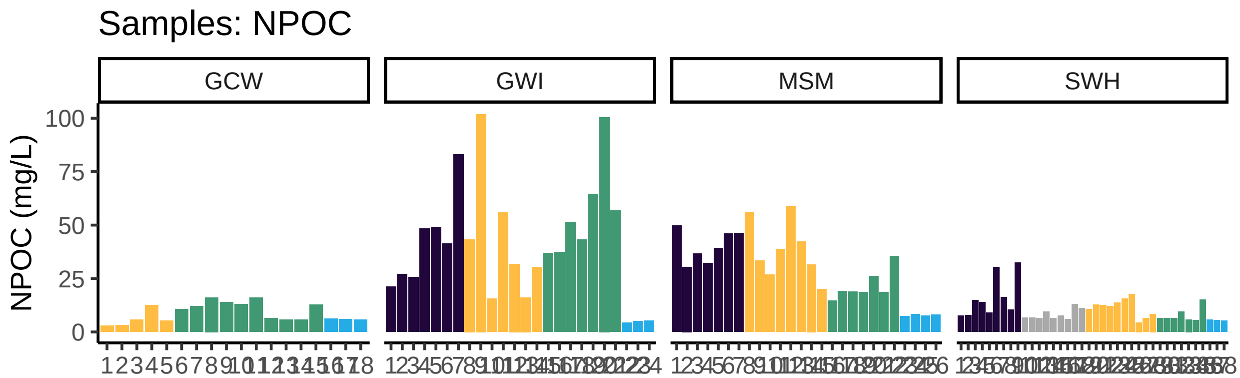
Sample Flagging - Are samples Within the range of the curve?

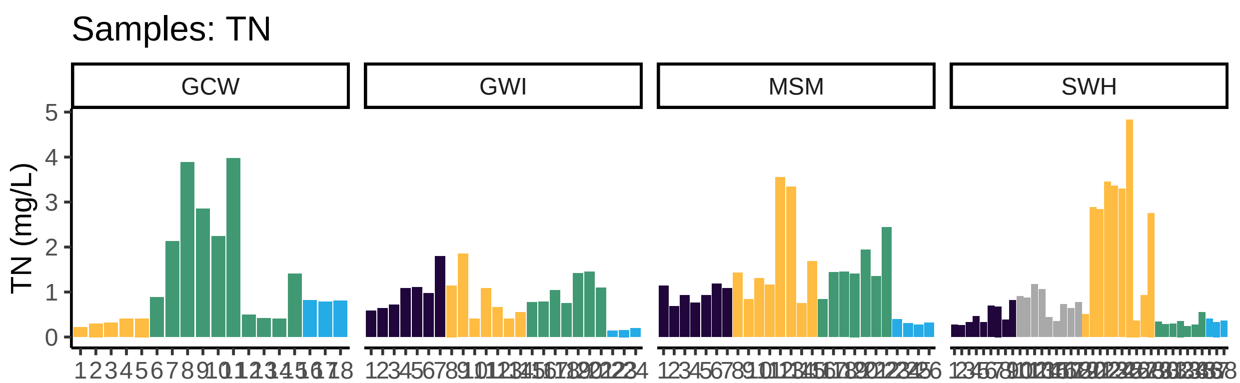
```
## Sample Flagging
```



Visualize Data by Plot

```
## Visualize Data

## Warning in rbind(c("202511", "SWH", "UP", "LysA", "10cm"), c("202511", "SWH", :
## number of columns of result is not a multiple of vector length (arg 1)
```





Convert data from mg/L to uMoles/L

Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata

## All sample IDs are present in metadata.
```

Export Processed Data

```
## Export Processed Data

## # A tibble: 6 x 21
##   Project      Region Site  Zone  Replicate Depth_cm Sample_ID  Year Month   Day
##   <chr>        <chr>  <chr> <chr> <chr>        <int> <chr>     <int> <int> <int>
## 1 COMPASS: Sy~ CB     SWH   UP    A               10 202511_S~  2025    11     5
## 2 COMPASS: Sy~ CB     SWH   UP    A               20 202511_S~  2025    11     5
## 3 COMPASS: Sy~ CB     SWH   UP    A               45 202511_S~  2025    11     5
## 4 COMPASS: Sy~ CB     SWH   UP    B               10 202511_S~  2025    11     5
## 5 COMPASS: Sy~ CB     SWH   UP    B               20 202511_S~  2025    11     5
## 6 COMPASS: Sy~ CB     SWH   UP    B               45 202511_S~  2025    11     5
## # i 11 more variables: Time <chr>, Time_Zone <chr>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

#end